

Comprehensive Report on OGD-Dublin Core-DCATv3 Metadata Mapping

Executive Summary

This report presents a detailed analysis of metadata mapping between India's Open Government Data (OGD) platform and two international standards: Dublin Core and DCATv3 (Data Catalog Vocabulary version 3). The mapping exercise reveals significant structural differences between these standards, particularly in how they handle subject classification, attribution, and the relationship between datasets and their distributions. While Dublin Core provides a simpler, flatter metadata structure, DCAT offers a more sophisticated three-tier model (Catalog-Dataset-Distribution) that better accommodates the complexity of government data publishing.

Introduction

National Informatics Centre (NIC) is working to standardize OGD dataset metadata across e-Governance portals, aiming for interoperability between the widely-used Dublin Core standard and the W3C's DCAT v3 (Data Catalog Vocabulary, version 3). Dublin Core provides 15 core metadata elements (e.g. Title, Subject, Date) for general resource description, whereas DCAT is an RDF vocabulary tailored for data catalogs and built upon many Dublin Core terms. In practice, DCAT reuses Dublin Core properties (via the DCMI Terms namespace) for common fields, and adds specialized classes (Dataset, Distribution, etc.) and properties for data catalog needs. This alignment is crucial for NIC, as it will allow Indian data portals (like Data.gov.in) to be compatible with international open data practices (the EU's DCAT-AP, the US's DCAT-US, etc.) while addressing local requirements.

Why align Dublin Core, OGD metadata and DCAT?

Dublin Core (DC) has been a simple, domain-agnostic standard for metadata, long adopted in government and library contexts. DCAT, on the other hand, is specifically designed for describing datasets in catalogs, enabling easier data discovery, federation, and exchange. DCAT is "grounded in the solid foundations of Dublin Core" - it leverages Dublin Core's familiar concepts (title, description, publisher, etc.) while extending them for dataset-specific needs (e.g. distributions, data services, versioning). By mapping DC elements to DCAT v3, NIC can ensure that metadata from various ministries and states remains interoperable (both internally and with global open data portals), without losing information during conversion. Additionally, understanding DCAT-AP (Europe's DCAT Application Profile) and DCAT-US (U.S. Federal

profile) is important, as they illustrate how DCAT is applied in different governance contexts. DCAT-AP (v3.0 aligned with W3C DCAT 3) is the standard in the EU for government data catalogs, adding requirements like controlled vocabularies for themes and mandated fields. DCAT-US (v3.0, recently updated from the Project Open Data schema) aligns with DCAT v3 while incorporating U.S.-specific fields (e.g. bureau codes, access levels) and geospatial metadata. For NIC, aligning with DCAT v3 means Indian open data can more easily integrate with such profiles if needed, and adopting best practices (FAIR data principles – Findable, Accessible, Interoperable, Reusable) already embodied in DCAT. The remainder of this report provides: (1) a detailed comparison of Dublin Core vs DCATv3 metadata elements; (2) a mapping of each Dublin Core element to DCATv3 (and notes on DCAT-AP and DCAT-US where relevant); (3) an analysis of interoperability issues in the Indian e-governance context; and (4) edge cases and recommendations for NIC's metadata standardization.

Part I: OGD to Dublin Core Mappings

1. Title Elements

OGD 'title' → Dublin Core 'Title' (dc:title or dcterms:title)

The OGD title field at both catalog and dataset levels maps directly to Dublin Core's Title element. This represents an exact match with no semantic loss. The mapping is straightforward as both expect free-text titles that uniquely identify the resource. Current OGD titles such as "Real time Air Quality Index from various locations" translate perfectly to Dublin Core.

OGD Alternative Title (currently absent) → Dublin Core 'Alternative' (dcterms:alternative)

OGD lacks an alternative title field, though AIKosh implements a "Short Description" that could serve this purpose. Dublin Core's Alternative element would accommodate abbreviated titles, acronyms, or translations. For instance, "Pradhan Mantri Awas Yojana Implementation Statistics" could have an alternative title "PMAY Stats". This represents a gap in OGD that could be augmented for AIKosh and OGD metadata to correspond perfectly.

2. Creator, Publisher, and Contributor Complexity

OGD 'cdos_state_ministry' → Dublin Core 'Creator' (dc:creator or dcterms:creator)

The mapping here is ambiguous because OGD uses **cdos_state_ministry** to represent what might be either a creator or publisher. Dublin Core's Creator element is meant for "an entity primarily responsible for making the content of the resource." In OGD's context, this field sometimes represents the data creator (like the Central Pollution Control Board for air quality data) and sometimes the ministry-department responsible for publishing the resource. The recommendation is to retain this mapping but clarify the semantic intent. In many cases, the dataset creator is either not known before publication or the same as the publisher. This metadata field is recommended to be optional and may take free text values.

OGD 'ministry_department' and 'state_department' → Dublin Core 'Publisher' (dc:publisher or dcterms:publisher)

These OGD fields map to Dublin Core's Publisher element, defined as "an entity responsible for making the resource available." However, OGD's structure with separate fields for ministry_department and state_department creates redundancy in Dublin Core, which has only a single Publisher element. The challenge is that Dublin Core cannot maintain the hierarchical relationship between ministry and department without additional structure. It's a mandatory field and maps to 'Author' in the AI Kosh metadata. So as a workaround we can modify the publisher field to accept an array containing ministry, department and other sub department provenance instead of the current form where it is a concatenated string.

Example:

Publisher = ["Ministry of Earth", "Forest and Climate Change", "CPCB", "Legal Cell"]

OGD 'contributor' fields → Dublin Core 'Contributor' (dc:contributor or dcterms:contributor)

OGD lacks explicit contributor fields for external entities. Since publisher field is already taken care of by **dcterms:publisher** field, **dcterms:contributor** could be added to OGD metadata to credit external contributors for the resource. Dublin Core's Contributor element would be valuable for crediting survey agencies, research institutions, or partner organizations involved in data creation. This represents an augmentation opportunity. In the data.gov.in platform 'Contributor' field corresponds to the publisher of the resource. It's recommended to keep it as an optional field with no limitations on what value this field can take.

3. Subject Classification Challenge

OGD 'keyword', 'sector', and 'sector_resource' → Dublin Core 'Subject' (dc:subject or dcterms:subject)

This mapping presents the most significant challenge in the OGD to Dublin Core transformation. Dublin Core has only a single Subject element that must accommodate three distinct OGD fields:

- **OGD 'keyword'**: Free-text tags like "Agricultural Marketing"
- **OGD 'sector'**: Controlled vocabulary from 36 predefined sectors like "Agriculture"
- **OGD 'sector_resource'**: Additional sector classifications at the resource level

The fundamental problem is that Dublin Core's Subject element does not distinguish between controlled and uncontrolled terms. Dublin Core allows duplication of metadata fields. Hence the 'Subject' keyword can in theory be used three times to map keyword, sector and sector_resource. When mapping these three OGD fields to Dublin Core Subject, we lose the semantic distinction between formal sector classifications and informal keywords. The recommendation is to keep using 'keyword', 'sector' and 'sector_resource' as is and add

dc:subject column to the database concatenating all three terms into a single compound word;
e.g

Subject: 'keyword-sector-sector_resource'= 'agriculture-agricultural_marketing-cotton_prices'.

4. Description and Abstract

OGD 'body' (catalog level) and 'note' (dataset level) → Dublin Core 'Description' (dc:description or dcterms:description)

OGD's approach to descriptions creates mapping challenges. Catalogs have a 'body' field while datasets have only a 'note' field. Both map to Dublin Core's Description element, but the semantic intent differs. The 'body' field contains actual descriptions while 'note' contains supplementary information. Dublin Core cannot maintain this distinction. Hence we suggest extending the Dublin Core vocabulary to add a field 'Note' that closely matches to the semantic meaning of 'note' at resource level.

Note [Extension to Dublin Core]: 'Note for the resource'

OGD Abstract (currently absent) / Short Description in AI Kosh → Dublin Core 'Abstract' (dcterms:abstract)

OGD currently does not have a dedicated abstract field. The 'note' field partially serves this purpose but conflates abstracts with general notes. Dublin Core's Abstract element, defined as "a summary of the resource," would be valuable for dataset summaries distinct from full descriptions. Short description in AIKosh can be mapped to the Dublin Core 'Abstract' field as an alternative to current mapping to 'Alternative Title'. When OGD datasets also have a short description field this would be backward compatible with Dublin Core metadata.

Abstract/Alternative Title: 'Short text for the dataset'

5. Date Elements

OGD 'created' → Dublin Core 'Created' (dcterms:created)

Direct mapping with exact semantic match. However, OGD's date format inconsistency (using formats like "4/8/2016" instead of ISO 8601) creates practical challenges. We recommend using standard date formats like DD-MM-YYYY or MM-DD-YYYY.

From D/M/YYYY to DD-MM-YYYY to MM-DD-YYYY

OGD 'published' → Dublin Core 'Issued' (dcterms:issued)

The OGD 'published' field maps to Dublin Core's Issued element. The semantic alignment is exact, representing the date of formal issuance. As mentioned above, we recommend using standard date formats.

From D/M/YYYY to DD-MM-YYYY to MM-DD-YYYY

OGD 'changed' → Dublin Core 'Modified' (dcterms:modified)

Direct mapping to Dublin Core's Modified element. The 'changed' field in OGD corresponds exactly to the last modification date concept in Dublin Core. We recommend using standard date formats.

From D/M/YYYY to DD-MM-YYYY to MM-DD-YYYY

OGD 'Duration of Date' → Dublin Core 'Date' (dc:date) or 'Temporal' (dcterms:temporal)

OGD's duration field, which captures temporal coverage like "01/01/2025-31/12/2025", maps awkwardly to Dublin Core. The basic Date element expects a single date, while dcterms:temporal can accommodate date ranges but lacks the structured format of DCAT's approach.

6. Format and Type Challenges

OGD 'file_format' → Dublin Core 'Format' (dc:format or dcterms:format)

This mapping is problematic because OGD stores format at the dataset level while Dublin Core expects it to describe the specific resource being accessed. OGD's practice of listing multiple formats like "CSV/XML/JSON/ODS" in a single field cannot be properly represented in Dublin Core without creating multiple resource records.

OGD 'resource_category' → Dublin Core 'Type' (dc:type or dcterms:type)

OGD's resource_category field (Dataset/Application) maps to Dublin Core's Type element. However, Dublin Core Type typically uses controlled vocabularies like the DCMI Type Vocabulary, which may not align with OGD's categories.

7. Identifier Challenges

OGD 'domain+node_alias' → Dublin Core 'Identifier' (dc:identifier or dcterms:identifier)

OGD uses the same URL pattern for multiple purposes, which Dublin Core cannot distinguish:

- As a unique identifier
- As a landing page URL

- As an access point

Dublin Core's Identifier element can accommodate the URL, but cannot maintain these semantic distinctions. So the recommendation is to keep the current 'domain' and 'node_alias' field and add a singular identifier field containing the URL of the landing page of the catalog or datasets. The landing page can serve as the unique identifier for that particular resource.

8. Source and Provenance

OGD 'field_reference_url' → Dublin Core 'Source' (dc:source or dcterms:source)

Direct mapping with good semantic alignment. The field_reference_url in OGD, which points to original data sources, maps cleanly to Dublin Core's Source element.

9. Language (Gap in OGD)

OGD Language (absent) → Dublin Core 'Language' (dc:language or dcterms:language)

OGD lacks language metadata despite India's multilingual context. Dublin Core's Language element uses ISO 639 codes and would be essential for indicating whether datasets are in Hindi, English, or regional languages. This would be useful in India's multilingual context.

10. Coverage Complexity

OGD 'field_asset_jurisdiction' → Dublin Core 'Spatial Coverage' (dcterms:spatial)

OGD's jurisdiction field maps to Dublin Core's spatial coverage, but Dublin Core's Coverage element historically conflated spatial and temporal aspects. Using dcterms:spatial is more precise but still cannot capture the hierarchical nature of Indian administrative divisions (Union→State→District→Block→Panchayat).

OGD 'Duration of Date' → Dublin Core 'Temporal Coverage' (dcterms:temporal)

Temporal coverage in OGD maps to Dublin Core's temporal elements, but loses structure. Dublin Core cannot maintain start and end dates as separate fields without additional encoding schemes. For date ranges, a slash (/) can separate the start and end dates, as in 2012-05-29/2012-05-31.

11. Rights and Licensing

OGD 'Released Under' → Dublin Core 'Rights' (dc:rights or dcterms:rights)

OGD's "Released Under" field (typically "NDSAP") maps to Dublin Core's Rights element. However, Dublin Core conflates license information with general rights statements, making it impossible to distinguish between copyright notices and reuse licenses.

OGD License (implicit GODL) → Dublin Core 'License' (dcterms:license)

While OGD assumes GODL licensing, it rarely makes this explicit. Dublin Core's License element (available only in dcterms, not basic dc) provides a place for license URLs.

Having controlled vocabulary of licensing agreements (CC 4.0, MIT , AGPL, GODL etc) will make it simpler to accommodate licensing agreements of other platforms (AIKosh, Bharat Data Platform etc).

OGD 'Access Type' → Dublin Core 'Access Rights' (dcterms:accessRights)

OGD's access classifications (Open/Registered/Restricted) map directly to Dublin Core's Access Rights element with good semantic alignment.

12. Relations (Largely Absent in OGD)

OGD Relationships → Dublin Core 'Relation' (dc:relation or dcterms:relation)

OGD lacks explicit relationship metadata, missing opportunities to use Dublin Core's rich relationship vocabulary. This metadata field is helpful for mapping relationships amongst datasets.

- dcterms:hasPart / dcterms:isPartOf (for composite datasets)
- dcterms:hasVersion / dcterms:isVersionOf (for dataset versions)
- dcterms:replaces / dcterms:isReplacedBy (for superseded datasets)
- dcterms:references / dcterms:isReferencedBy (for linked resources)

This field is useful for consolidating or bifurcating multiple datasets or catalogs and maintaining the semantic meaning.

13. Fields Without Dublin Core Equivalents

Several OGD fields have no Dublin Core equivalent. Dublin Core and DCAT does not recommend having boolean or numerical values as metadata.

- **ogdp_view_count**: View statistics
- **ogdp_download_count**: Download statistics
- **is_visualized**: Visualization availability flag
- **is_rated**: Rating capability flag
- **field_high_value_dataset**: High-value dataset indicator
- **field_from_api**: API availability flag
- **api_request_count**: API usage statistics

These represent metadata that Dublin Core's document-centric model cannot accommodate, as they are system-specific, boolean, or statistical values.

Part II: OGD to DCAT v3 Mappings

DCAT provides a more sophisticated model with three distinct classes: Catalog, Dataset, and Distribution. This allows for better representation of OGD's structure.

1. Catalog-Level Mappings

OGD Catalog 'title' → DCAT 'dcterms:title'

Direct mapping with exact semantic match. DCAT reuses Dublin Core terms where appropriate.

Example: **dcterms:title = 'Real time Air Quality Index from various locations'**

OGD 'field_ministry_department' → DCAT 'dcterms:publisher' and 'prov:qualifiedAttribution'

DCAT's advantage over Dublin Core becomes apparent here. While the basic publisher maps to dcterms:publisher, DCAT can use prov:qualifiedAttribution to specify:

- The ministry as publisher
- The department as a contributor with a specific role
- Specific responsibilities

This allows representing complex organizational hierarchies that Dublin Core cannot handle. Publisher is mandatory in DCAT, specific departments can be mapped in a hierarchical manner following the DCAT 'dcterms:qualifiedAttribution'.

Example:

Publisher = **"Ministry of Environment Forest and Climate Change"**

Provenance = [{ **Qualified Attribution = 'Central Pollution Control Board',**

Role: Data Collector } → {Qualified Attribution = 'Legal Cell'

Role: Data Curator/ Data Steward}]

Defining specific roles for organizations in the provenance chain is optional. This hierarchical structure makes it easier to track data publisher information in one place.

OGD 'field_sector' → DCAT 'dcat:theme'

Unlike Dublin Core's single Subject element, DCAT explicitly separates themes (controlled vocabulary) from keywords (free text). OGD's **sector** field maps cleanly to dcat:theme. This is a mandatory field recommended to take only pre-defined values(controlled vocabularies).

Example:

Theme: sector= 'Agriculture'

In data.gov.in some datasets can have multiple sectors. e.g

For the dataset resource '**Real time air quality index for various locations**' has multiple sectors populated in the 'sector' metadata field = 'Environment and Forest;Industrial Air Pollution;Residential Air Pollution;Vehicular Air Pollution'.

There are a couple of ways this issue can be resolved.

1. Store themes field in an array with multiple sectors in an array instead of the single concatenated string based approach.
 - a. Themes:Sectors = ["Environment and Forest", "Industrial Air Pollution", "Residential Air Pollution", "Vehicular Air Pollution"]
2. Normalized approach: create separate "themes" table and add each sector as a row.

Main table

dataset_id: 123

title: "Real time air quality index for various locations"

Themes table

dataset_id | theme

123 | Environment and Forest

123 | Industrial Air Pollution

123 | Residential Air Pollution

OGD 'keywords' → DCAT 'dcat:keyword'

Separate from themes. Multiple keywords can be grouped together in a single array (as explained above) or separate normalized keywords table. This field is recommended to be mandatory and no limitation on the text value this metadata field can take.

Example:

Keywords: keyword = ['Refinery', 'Oil']

Or **dataset_id | keyword**

123 | Refinery

123 | Oil

OGD 'group' → DCAT 'dcat:subject'

The catalog-level grouping in OGD can map to DCAT's subject property, distinct from themes and keywords.

Example:

Group: dcat:subject= ["Refinery"]

OGD 'body' → DCAT 'dcterms:description'

Direct mapping for catalog descriptions. DCAT has direct mapping from 'body' to 'dcterms:description'. Changing the nomenclature from body to description might be taken up.

Example

Body:description = "The catalog contains monthly data on crude oil processed by refineries."

OGD 'field_ds_govt_type' → DCAT-India extension 'dcatin:jurisdictionLevel'

DCAT allows extensions for local requirements. The proposed **dcatin:jurisdictionLevel** would capture India's complex federal structure (Central/State/District/Block/Panchayat) which neither Dublin Core nor standard DCAT can represent. Controlled vocabulary is suggested here in reference to the level of government (Central/State). This structure would be backward compatible if NIC allows local governments to publish catalogs and resources on the platform.

Example:

Field_ds_govt_type: JurisdictionLevel="District"

OGD 'high_value_dataset' → DCAT-AP 'dcatap:hvdCategory'

The European DCAT Application Profile provides a high-value dataset categorization using SKOS (SKOS, or Simple Knowledge Organization System) controlled vocabularies. OGD's

binary indicator would map to specific categories in the Indian context as DCAT does not recommend using Boolean values(True/False). It is recommended to lay down specific guidelines on how to tag a specific resource to high value categories. The current task of high value dataset tagging by CDOs could be simplified if specific HVD criteria guidelines are laid down. (Precedent EU's [open data and the re-use of public sector information](#)).

Using a weightage based scoring model to tag high value datasets could result in false positives and negatives i.e miscategorization (and exclusion) of resources in HVD category. Black-box based mathematical models take away the responsibility of HVD tagging from CDOs.

Example:

High_value_dataset: High Value Dataset Category

OGD 'Released Under' → DCAT-AP 'dcatap:applicableLegislation' or DCAT-India 'dcatin:applicableLegislation'

DCAT extensions can capture the legislative framework (NDSAP) under which data is released. Following DCAT-AP (European Union) vocabulary, the act under which the dataset/catalog is released could be mapped to applicable legislation. For this field extension to DCAT standard dcatin:applicableLegislation is recommended taking into account NDSAP and other future regulations.

Example:

dcatin:applicableLegislation = 'National Data Sharing and Accessibility Policy'.

For other platforms, the respective act under which the resource has been opened up for public consumption could be signified. This is a mandatory requirement with a controlled vocabulary.

2. Dataset-Level Mappings

OGD Dataset 'title' → DCAT 'dcterms:title'

Direct mapping, identical to catalog level.

OGD 'cdos_state_ministry' → DCAT 'dcterms:creator' or 'prov:qualifiedAttribution'

DCAT can maintain the distinction between simple creator attribution and complex multi-organizational involvement through qualified attribution as most of the datasets have multiple ministerial departments for creating/publishing the dataset.

This allows representing complex organizational hierarchies that Dublin Core cannot handle. Publisher is mandatory in DCAT, specific departments can be mapped in a hierarchical manner following the DCAT 'dcterms:qualifiedAttribution'.

Example:

Creator = “**Ministry of Environment Forest and Climate Change**”

Provenance = [{ **Qualified Attribution = ‘Central Pollution Control Board’,**
Role: Data Collector } → {Qualified Attribution = ‘Legal Cell’
Role: Data Curator/ Data Steward}}]

Defining specific roles for organizations in the provenance chain is optional. This hierarchical structure makes it easier to track data publisher information in one place. This field is kept optional and can take free text as values.

OGD 'sector' and 'sector_resource' → DCAT 'dcat:theme'

DCAT handles multiple theme assignments elegantly. The "All" option in OGD sectors should be deprecated, with explicit sector selection required – i.e instead of checkboxing ‘All’ the publisher can auto select all sectors in the dropdown.

mandatory field recommended to take only pre-defined values(controlled vocabularies).

Example:

Theme: sector= ‘Agriculture’

In data.gov.in some datasets can have multiple sectors. e.g

For the dataset resource ‘**Real time air quality index for various locations**’ has multiple sectors populated in the ‘sector’ metadata field = ‘Environment and Forest;Industrial Air Pollution;Residential Air Pollution;Vehicular Air Pollution’.

There are a couple of ways this issue can be resolved.

3. Store themes field in an array with multiple sectors in an array instead of the single concatenated string based approach.
 - a. Themes:Sectors = [“Environment and Forest”, “Industrial Air Pollution” ,”Residential Air Pollution” , “Vehicular Air Pollution”]
4. Normalized approach: create separate “themes” table and add each sector as a row.

Main table

dataset_id: 123

title: "Real time air quality index for various locations"

Themes table

dataset_id | theme

123 | Environment and Forest

123 | Industrial Air Pollution

123 | Residential Air Pollution

OGD 'keyword' → DCAT 'dcat:keyword'

Maintained separately from themes, recommend to follow similar methodology as explained above for catalog metadata.

This field is recommended to be mandatory and no limitation on the text value this metadata field can take.

Example:

Keywords: keyword = ['Refinery', 'Oil']

Or **dataset_id | keyword**

123 | Refinery

123 | Oil

OGD 'note' → DCAT 'dcatin:note'

In line with Dublin Core recommendation, to include the 'note' field in the proposed metadata standard, we propose an extension named 'dcatin:note'. This would be an optional parameter with no limitation on what value it can take.

Example:

Note: dcatin:note="This dataset does not contain FY-23 details"

OGD 'ministry_department' and 'state_department' → DCAT 'dcterms:publisher' with 'prov:qualifiedAttribution'

DCAT can represent the full organizational hierarchy through qualified attribution, maintaining relationships between ministries and their departments.

Example:

Publisher:dcterms:publisher = **"Ministry of Environment Forest and Climate Change"**

Provenance = [{ **Qualified Attribution = 'Central Pollution Control Board',**
Role: Data Collector } → {Qualified Attribution = 'Legal Cell'
Role: Data Curator/ Data Steward}]

Defining specific roles for organizations in the provenance chain is optional. This hierarchical structure makes it easier to track data publisher information in one place. This field is kept mandatory and can take free text as values.

OGD Date Fields → DCAT Temporal Properties

- 'created' → 'dcterms:created'
- 'published' → 'dcterms:issued'
- 'changed' → 'dcterms:modified'

All require format standardization to ISO 8601.

We recommend using standard data formats **DD-MM-YYYY or MM-DD-YYYY** as opposed to the current format of M/DD/YYYY format. Published and modified are mandatory fields taking date type values only.

OGD 'Duration of Date' → DCAT 'dcterms:temporal'

DCAT provides structured temporal coverage using dcterms:PeriodOfTime with explicit dcat:startDate and dcat:endDate properties, far superior to Dublin Core's approach. This date range for which the dataset is relevant e.g Financial year, data ranges, Monsoon season, Hourly etc. This field is present in the dataset create form.

Example:

```
Duration of Date (from- to):dcterms:temporal [ a dcterms:PeriodOfTime ;  
    dcat:startDate "2016-03-04"^^xsd:date ;  
    dcat:endDate "2018-08-05"^^xsd:date ;  
]
```

OGD 'resource_category' → DCAT 'dcterms:type'

The dataset type is somewhat redundant in DCAT since the class itself indicates a dataset, but can be retained for subtype classification in OGD context. Dataset type implied by DCAT class; retain if subtype or classification needed. E.g if NIC required separation amongst Datasets, Documents, Applications, Charts etc

Example:

resource_category:dcterms:type=Dataset/Application/Database

OGD 'file_format' and 'field_resource_type' → DCAT Distribution level

These fields should not exist at the dataset level in DCAT. They must be migrated to distribution-level metadata. While the legibility of resource type could be improved if numerical values are replaced with their string values.

OGD 'domain+node_alias' → Multiple DCAT Properties

- 'dcterms:identifier': For unique identification
- 'dcat:landingPage': For the human-readable landing page
- 'dcat:accessURL': Moved to distribution level if data is consumed via APIs or other web services.

DCAT properly separates these three distinct concepts.

OGD 'field_reference_url' → DCAT 'dcterms:source'

Direct mapping for indicating original data sources. This is an optional field with free text. It indicates derived resources from any report, website, document etc.

Example:

field_reference_url:dcterms:source="<https://sansad.in/rs/questions/questions-and-answers>**"**

For Rajyasabha session question-answers.

OGD 'frequency' → DCAT 'dct:accrualPeriodicity'

Update frequency maps directly, though DCAT prefers controlled vocabulary from the EU frequency vocabulary. In the OGD context we recommend this field to be controlled vocabulary. This field is useful for datasets published in a series (GDP series, CPI etc).

OGD 'granularity' → DCAT 'dcat:temporalResolution'

Temporal granularity (hourly, daily, monthly) maps to DCAT's temporal resolution property.

OGD 'Access Type' → DCAT 'dcterms:accessRights'

Direct mapping for access restrictions.

Example:

Access Type (on dataset creation page):dcterms:accessRight="Open/Registered/Restricted"

OGD Missing Language → DCAT 'dcterms:language'

Though this field is present in OGD's internal portal, it's not clear how it's used in the platform. It's recommended to include this field for vernacular datasets as well as metadata translation to other Indian languages.

OGD Missing Spatial Coverage → DCAT 'dcterms:spatial'

Currently inferred from domain structure, must be made explicit with proper geographic identifiers or coordinates. Currently this field is inferred from the 'domain' or 'Asset Jurisdiction' field. It's recommended to add an explicit spatial field according to DCAT convention.

Example:

If administrative hierarchies are required to be shown then store the geographies in the array.

Asset Jurisdiction (on the data.gov.in platform): dcterms:spatial = ["Maharashtra", "Pune"]

3. Distribution-Level Mappings

This represents the most significant structural change from OGD to DCAT.

New Distribution Entity Requirements

DCAT requires creating separate distribution entities for each format/access method. A single OGD dataset with "CSV/XML/JSON" formats becomes three DCAT distributions.

OGD 'datafile' → DCAT 'dcat:downloadURL'

The URL of the downloadable file in a given format. Direct download links map to downloadURL at the distribution level. The format is indicated by the distribution's dcterms:format

OGD 'file_format' → DCAT 'dcat:mediaType'

Format information moves from dataset to distribution level, using IANA media types (text/csv, application/json) instead of informal format names.

OGD 'file_size' → DCAT 'dcat:byteSize'

File size in bytes maps directly at distribution level. The size in bytes can be approximated (as a non-negative integer) when the precise size is not known.

OGD 'datafile_url' (API endpoint) → DCAT 'dcat:DataService'

API endpoints require a separate DataService entity with:

- 'dcat:endpointURL': The API base URL
- 'dcat:endpointDescription': API documentation
- 'dcat:servesDataset': Link back to dataset

4. Ambiguous Mappings and Gaps

Multi-Sector Datasets (Rajya Sabha Questions)

OGD's 'sector_resource' field attempts to handle datasets spanning multiple sectors. DCAT handles this better through multiple theme assignments, but the practice of selecting "All" sectors must be deprecated. There is no option of selecting all sectors in AI Kosh

Attribution Complexity

Many OGD datasets involve multiple organizations (creator, publisher, data collector, validator). Dublin Core cannot represent this complexity, while DCAT's qualified attribution provides a solution through:

```
prov:qualifiedAttribution [  
  a prov:Attribution ;  
  prov:agent <organization-uri> ;  
  dcat:hadRole <role-uri> ;  
  prov:generatedAtTime "date" ;  
]
```

Boolean and Statistical Metadata

Fields like `is_visualized`, `is_rated`, `ogdp_view_count` have no place in either Dublin Core or DCAT. Options include:

- Deprecation from core metadata
- Implementation as extension properties
- Migration to analytics systems

Composite Datasets

OGD cannot represent relationships between parent datasets and their components. DCAT provides:

- `dcterms:hasPart`
- `dcterms:isPartOf`

Essential for hierarchical data organization.

5. India-Specific Extensions Required

Jurisdictional Hierarchy (`dcatin:jurisdictionLevel`)

Must accommodate:

- Union Government
- State Governments (28 states)
- Union Territories (8)
- Districts
- Blocks/Tehsils (6,000+)
- Panchayats (250,000+)
- Urban Local Bodies

Applicable Legislation (`dcatin:applicableLegislation`)

Must reference:

- National Data Sharing and Accessibility Policy (NDSAP)
- State-specific policies
- Sector regulations

High-Value Dataset Categories (`dcatin:hvdCategory`)

Using the six Indian thematic categories

1. Governance and Service Delivery
2. Economic and Business
3. Infrastructure and Connectivity

- 4. Environment, Climate and Agriculture
- 5. Health, Education and Human Development
- 6. Socio-Economic and Demographic

Part III: Critical Observations and Recommendations

Fundamental Structural Misalignments

The core challenge in mapping OGD to both Dublin Core and DCAT stems from OGD's conflation of conceptual levels. OGD treats datasets and their distributions as a single entity, stores format information at the wrong level, and cannot properly represent multi-organizational attribution or complex relationships between datasets.

Dublin Core, being document-centric, cannot adequately represent the complexity of government data publishing. Its single-level structure and limited relationship vocabulary make it unsuitable as a primary metadata standard for OGD, though it remains valuable for basic interoperability.

DCAT's three-tier model (Catalog-Dataset-Distribution) aligns better with the reality of government data publishing. However, implementing DCAT requires fundamental restructuring of OGD's data model, not merely field mapping.

Priority Gaps Requiring Immediate Attention

The most critical gaps that affect data usability and must be addressed immediately include:

Distribution Separation: The current practice of listing multiple formats in a single field must end. Each format requires a separate distribution entity with its own metadata including media type, byte size, and access URL.

Example: Instead of listing the format like this: json/csv the format can be listed separately as its own entity for.

dataset_title	distribution_title	accessURL	mediaType	byteSize	frequency	granularity
Variety-wise Daily Market Prices Data of Commodity	CSV download link	https://example.com/data.csv	text/csv	14645	Daily	Daily

Variety-wise Daily Market Prices Data of Commodity	JSON API endpoint	https://example.com/data.json	application/json	23453	Daily	Daily
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Identifier Disambiguation: The same URL cannot serve as identifier, landing page, and access URL. These must be separated with clear semantic distinctions.

Attribution Complexity: The current Contributor field cannot represent the reality of multi-organizational dataset creation. Implementation of qualified attribution is essential.

Temporal and Spatial Coverage: These must be explicit, machine-readable, and standardized rather than inferred or free-text.

Language Metadata: In a multilingual nation, the absence of language metadata is a critical gap that must be filled.

Recommendations for Mapping Strategy

For Dublin Core compatibility, maintain simple mappings for basic interoperability but acknowledge limitations. Use Dublin Core for baseline metadata exchange while recognizing that complex government data relationships cannot be fully represented.

For DCAT implementation, adopt a phased approach starting with distribution separation, then attribution enhancement, followed by relationship mapping and India-specific extensions.

For fields without standard mappings (view counts, ratings, visualization flags), either deprecate from core metadata or implement as properly namespaced extensions, but do not attempt to force them into document-centric models.

Special Considerations for Edge Cases

Multilingual Datasets: Beyond simple language tags, need metadata for translation completeness, script variations (Devanagari vs Roman), and regional dialect indicators.

Composite Datasets: Rajya Sabha question datasets that span multiple sectors require careful modeling with proper part-whole relationships and clear sector attribution at appropriate levels.

This comprehensive mapping analysis reveals that while OGD has established a functional metadata structure, enhancements are needed to meet international standards and serve India's complex requirements. The path forward requires not just field mapping but fundamental structural changes, particularly in distribution management and attribution modeling.

